

SVM

- * promising method for classification of both linear and non linear data
- * uses non linear mapping to transform the original training data into higher dimension
- * With in the new dimension, it searches for linear optimal separating hyperplane.
- * SVM finds this hyperplane using support vectors and margins.
- training time is extremely slow, they are highly accurate
- much ^{less} prone to overfitting than other methods
- used for prediction as well as classification.
- used in handwritten digit recognition, object recognition, speaker identification
- time scales prediction tasks

SVM

equation of hyper plane

$$W^T x = 0$$

$$W \cdot x + b = 0$$

given hyper plane H_0 separating data

satisfying $W \cdot x + b = 0$

we can select two hyper-planes H_1 & H_2

$$W \cdot x + b = \delta$$

$$\& W \cdot x + b = -\delta$$

so H_0 is equal distance from H_1 & H_2

$$\delta = 1 \quad W \cdot x + b = 1$$

$$\& W \cdot x + b = -1$$

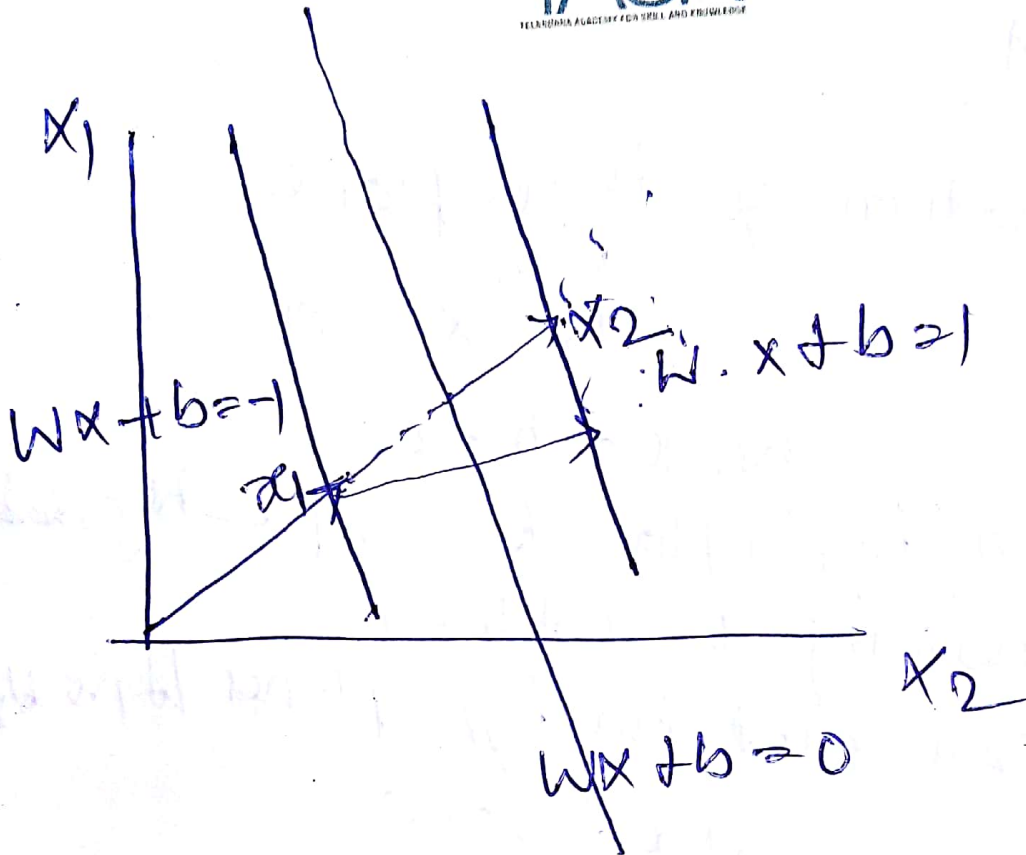
for each vector x_j

$$W \cdot x_j + b \geq 1 \quad \text{for } x_j = 1$$

$$W \cdot x_i + b \leq -1 \quad \text{for } x_i = -1$$

$$y_j (W \cdot x_j + b) \geq y_j (1)$$

$$y_i (W \cdot x_i + b) \geq -1$$



max $\frac{2}{||W||}$

$$Wx_2 + b = 1$$

$$Wx_1 + b = -1$$

$$Wx_2 + b - Wx_1 - b = 1 - (-1)$$

$$Wx_2 - Wx_1 = 2$$

$$\frac{W}{||W||} (x_2 - x_1) = \frac{2}{||W||}$$

we find W and b
min $\frac{1}{2} ||W||^2$

$$d(x^T) = \sum_{i=1}^l y_i \alpha_i x_i x^T + b_0$$

y_i is the class label of
Support vector x_i ,

x^T is a tuple vector
test tuple

α_i and b_0 are numeric values
that were determined by the
optimization

l support vectors